

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Determine the Undetermined

1. Revisiting the Homogeneous Case

We have been exploring a *2nd-order linear ODE with constant coefficients* of the form

$$y'' + py' + qy = f(t)$$

where $f(t) = 0$ (homogeneous) or $f(t) \neq 0$ (non-homogeneous). The general solution is of the form

$$y = y_h + y_p$$

where y_h is the homogeneous solution and y_p is the particular solution –both are functions of t . We solved for y_h using the characteristic equation. Now, we need to solve for y_p .

Consider the 2nd-order linear ODE with constant coefficients:

$$y'' + 2y' - 3y = f(t)$$

where $f(t)$ is some function.

a. Determine the roots of characteristic equation for the homogeneous case, $f(t) = 0$.

c. Suppose $f(t) = 6$, a constant function. Let $y_p = A$ for some constant A . Use the verification process to determine the value of A . Then, write y_p .

b. Write the homogeneous solution y_h .

d. Solve for the general solution $y = y_h + y_p$.

2. The Method of Undetermined Coefficients

The goal of the method of *undetermined coefficients* is to find a suitable y_p so that it satisfies the ODE.

- If $f(t) = 0$, then $y_p = 0$.
- If $f(t) \neq 0$ then we need to strategically pick an *Ansatz* (or a trial solution) for the particular solution.

Consider the same ODE from Problem (1), $y'' + 2y' - 3y = f(t)$ where $f(t)$ is some function.

- Let $f(t) = 2t - 1$. Here, we consider the Ansatz $y_p = A_1t + A_0$ because the structure of $f(t)$ is a linear function. The following tasks leads you on how to solve for y_p .

Task	Show work here
Take derivatives of y_p up to the highest order of the given ODE. _____	_____
Substitute y_p and its derivatives into the ODE. _____	_____
Match the coefficients of like terms from the left-hand side to the right hand-side of the equation. _____	_____
Solve for the unknowns. _____	_____
Write the particular solution y_p .	

- Let $f(t) = e^{-3t}$. Use the same process in Part (a) to determine the particular solution, y_p .

- A key step in the method of undetermined coefficients is to write the Ansatz for y_p . For each $f(t)$, use the table of trial solutions to write a suitable Ansatz. Do not solve for the particular solution.

- $f(t) = e^t (t^2 + 1)$

- $f(t) = e^{-2t} \cos(2t)$